

Linear constraints

Polyhedron

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Optimization: principles and algorithms



Polyhedron

When the set of constraints is defined by linear (affine) functions, the set is called a polyhedron.

Polyhedron

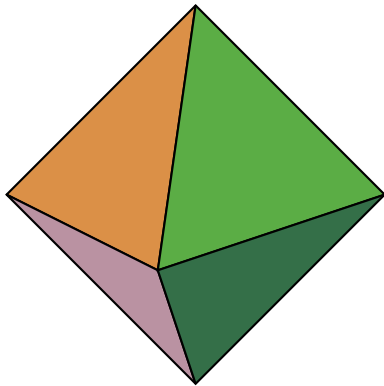
$$\{x \in \mathbb{R}^n \mid Ax \leq b\}$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$.

Polyhedron

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Convex set

A polyhedron $\mathcal{P}$ is a convex set

For all $x, y \in \mathcal{P}$, for all $0 \leq \lambda \leq 1$,

$$\lambda x + (1 - \lambda)y \in \mathcal{P}.$$

Polyhedron representations

$$A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m:$$

Canonical form

$$\{x \in \mathbb{R}^n \mid Ax \leq b\}$$

Standard form

$$\{x \in \mathbb{R}^n \mid Ax = b, x \geq 0\}$$

Geometric interpretation

- ▶ Canonical form:

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- ▶ Active constraints:

$$a_j^T x = b_j \iff x_j^s = 0.$$

Summary

Polyhedron: intersection of half-spaces.

Geometric object characterizing linear constraints of an optimization problem.

For linear algebra manipulation, standard form is preferred.